CLASS 8 SCIENCE

Elements, Atoms and Symbols of Elements

1. Matter, Atoms and Elements

Matter: Anything that has mass and occupies space is called matter. Matter is made up of tiny particles.

Atom: An atom is the smallest basic building block of an element. Atoms are extremely tiny.

Element: An element is a pure substance made up of only one kind of atom. It cannot be broken down into simpler substances by ordinary chemical methods.

Examples of elements: Iron, copper, gold, silver, oxygen, hydrogen and carbon are elements.

Atom vs element: An atom is one tiny particle; an element is a pure substance made of only one type of atom. For example, one gold atom is a particle of the element gold.

2. Dalton's Atomic Theory — Key Points

1. All matter is made up of tiny particles called atoms.

2. Atoms of the same element have similar properties. In Dalton's original theory, atoms of different elements were considered different.

3. Atoms of different elements can combine in fixed, whole-number ratios to form compounds.

4. Chemical reactions involve the rearrangement of atoms.

5. In a chemical reaction, atoms are not created or destroyed; they are rearranged. (This is the simplified school-level statement of the theory.)

3. Symbols of Elements

Meaning: A chemical symbol is a short form used to represent an element.

Writing symbols: A symbol has one or two letters. The first letter is always capital; the second letter, if present, is lowercase.

Examples: Hydrogen = H

Oxygen = O

Nitrogen = N

Carbon = C

Calcium = Ca

Chlorine = Cl

Magnesium = Mg.

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Remember: Co means cobalt, while CO means carbon monoxide (a compound). Letter case matters.

4. Symbols from English Names

One-letter symbols: Hydrogen — H; Boron — B; Carbon — C; Nitrogen — N; Oxygen — O; Fluorine — F; Phosphorus — P; Sulphur — S.

Two-letter symbols: Aluminium — Al; Argon — Ar; Barium — Ba; Beryllium — Be; Cobalt — Co; Helium — He; Iodine — I; Lithium — Li; Manganese — Mn; Neon — Ne.

5. Symbols from Latin Names

Why Latin names?: Some symbols come from the Latin names of elements rather than their English names.

Important examples: Sodium — Natrium — Na; Potassium — Kalium — K; Iron — Ferrum — Fe; Copper — Cuprum — Cu; Silver — Argentum — Ag; Gold — Aurum — Au; Mercury — Hydrargyrum — Hg; Lead — Plumbum — Pb; Tin — Stannum — Sn.

6. Easy-to-Learn Symbol Table

Element	Symbol
Sodium	Na
Potassium	K
Copper	Cu
Silver	Ag
Gold	Au
Mercury	Hg
Iron	Fe
Lead	Pb
Tin	Sn
Calcium	Ca
Magnesium	Mg
Aluminium	Al

7. Elements and Compounds

Element: Contains only one kind of atom. Examples: copper, carbon and helium.

Compound: A substance formed when atoms of two or more different elements combine chemically in a fixed ratio. Examples: water and carbon dioxide.

Water: Water is made of hydrogen and oxygen. Its formula is H_2O.

Carbon dioxide: Carbon dioxide is made of carbon and oxygen. Its formula is CO_2.

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8. Experiment: Passing Electricity Through Water

Aim: To observe what happens when electricity passes through water.

Materials: Water, a beaker, two test tubes, a 9-volt battery, connecting wires/electrodes, a few drops of dilute sulphuric acid (only under teacher supervision), and a test for the gases.

Method (summary): Fill a beaker about two-thirds with water. Add a few drops of dilute sulphuric acid only as directed by the teacher. Fill two test tubes with the same water and invert them in the beaker. Place the electrodes connected to the battery under the openings of the test tubes. Wait and observe gas bubbles collecting in the tubes.

Observations: Gas bubbles form at both electrodes. A larger volume of gas is generally collected at one electrode than the other; in the ideal experiment, hydrogen is produced in about twice the volume of oxygen.

Testing the gases: Hydrogen gives a 'pop' sound when a burning splint is brought near it. Oxygen makes a glowing splint burn brighter.

Conclusion: Water can be decomposed by electricity into hydrogen and oxygen. Water is a compound, not an element.

Safety: This is a teacher-supervised demonstration. Do not handle dilute acid, batteries, flames or collected gases without a teacher. Never try to ignite a large quantity of gas.

9. Important Definitions

Atom: The tiny basic building block of an element.

Element: A pure substance containing only one kind of atom.

Chemical symbol: A short letter form used to represent an element.

Compound: A pure substance made when two or more elements combine chemically in a fixed ratio.

Chemical reaction: A process in which atoms are rearranged to form new substances.

10. Quick Revision Points

1. An element contains only one kind of atom.
2. A chemical symbol starts with a capital letter.
3. If a symbol has two letters, the second letter is lowercase.
4. Some element symbols come from Latin names.
5. Water contains hydrogen and oxygen: H_2O.
6. Electrolysis of water produces hydrogen and oxygen.
7. Hydrogen gives a pop sound; oxygen supports burning.
8. Chemical reactions rearrange atoms.

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A. Multiple-Choice Questions (Choose one)

1. Which of the following is an element?

(a) Water (b) Copper (c) Carbon dioxide (d) Salt

2. What is the chemical symbol for sodium?

(a) So (b) S (c) Na (d) N

3. Which statement about a chemical symbol is correct?

(a) Both letters are always capital (b) First letter is capital (c) First letter is lowercase (d) Symbols always have three letters

4. Which element has the symbol Fe?

(a) Fluorine (b) Iron (c) Gold (d) Silver

5. Which gas gives a pop sound when tested with a burning splint?

(a) Oxygen (b) Nitrogen (c) Hydrogen (d) Carbon dioxide

6. Water is made up of:

(a) Hydrogen and oxygen (b) Copper and oxygen (c) Carbon and hydrogen (d) Sodium and chlorine

7. The symbol Ag stands for:

(a) Argon (b) Silver (c) Gold (d) Aluminium

8. What happens to atoms during a chemical reaction?

(a) They vanish (b) They are rearranged (c) They become nothing (d) They are always created

B. Fill in the blanks

1. The smallest basic building block of an element is an _____.
2. The chemical symbol for potassium is _____.
3. The symbol for gold is _____.
4. A pure substance made of only one kind of atom is called an _____.
5. Water has the chemical formula _____.
6. The two gases formed when electricity passes through water are _____ and _____.

C. True or False

1. The symbol for cobalt is CO.
2. The first letter of a chemical symbol is always capital.
3. Water is an element.
4. Hydrogen gives a pop sound in the burning-splint test.

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5. Some chemical symbols are based on Latin names.

D. Short-answer questions

1. What is an atom?
2. What is an element? Give two examples.
3. Write any four rules or facts about chemical symbols.
4. Why is the symbol for sodium Na and not S?
5. Name the two gases obtained by passing electricity through water. How can they be tested?
6. Write the symbols for iron, copper, silver, gold, mercury and potassium.
7. State any three points of Dalton's atomic theory.
8. Explain the difference between an atom and an element in simple words.

E. Application / thinking questions

1. A student writes the symbol for calcium as CA. Correct it and state the rule.
2. A student says that water is an element because it looks uniform. Explain why this is incorrect.
3. Why are bubbles seen at both electrodes during the electrolysis of water?
4. A gas collected in a test tube makes a glowing splint burn brighter. Which gas is likely present?

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